Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Mathematics

SS Mathematics **Trinity Term 2020**
SS Theoretical Physics

MAU44400 Quantum Field Theory

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Instructions that apply to all take-home exams

1. This is an open-book exam. You are allowed to use your class notes, textbooks and any material that is available through the internet. However, you are not allowed to seek help from others and you are not allowed to post questions on online forums such as Stack Exchange.

2. If you have any questions about the content of this exam, you may seek clarification from the lecturer using the e-mail address provided. You are not allowed to discuss this exam with others.

3. Solutions must be submitted through Blackboard by 12pm noon on Monday, May 4th. You must submit a single pdf file for each exam separately and sign the following declaration in each case.

Plagiarism declaration: I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar which are available through https://www.tcd.ie/calendar.

Signature: _____

Additional instructions for this particular exam

Attempt all questions. All questions have equal weight.

1. Scalar QED: 25 marks

Consider the EM fields coupled to charged matter, namely a complex scalar field $\Phi(x); \Phi(x) \neq \Phi^*(x)$ of electric charge $e \neq 0$. Altogether, with electromagnetic field, the Lagrangian is:

$$\mathcal{L} = \mathcal{L}_{EM} + \mathcal{L}_{mat} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + D^\mu\Phi^* D_\mu\Phi - m^2\Phi^*\Phi$$

$$\mathcal{L}_{EM} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu}$$

$$\mathcal{L}_{mat} = D^\mu\Phi^* D_\mu\Phi - m^2\Phi^*\Phi$$

where

$$D_\mu\Phi = (\partial_\mu + ieA_\mu)\Phi, \qquad D_\mu\Phi^* = (\partial_\mu - ieA_\mu)\Phi^*$$

are covariant derivatives. Gauge transformations are:

$$\Phi \to e^{i\alpha}\Phi, \qquad \Phi^* \to e^{-i\alpha}\Phi^*, \qquad A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$$

a) 5 marks: Write down the equation of motion for all fields in a covariant form (using covariant derivatives).

b) 5 marks: Write down the electric current which by definition is:

$$J^\mu = -\frac{\partial\mathcal{L}}{\partial A_\mu}$$

in a manifestly gauge-invariant form and verify its conservation:

$$\partial_\mu J^\mu = 0$$

(as long as the scalar fields satisfy their equations of motion).

c) 5 marks: Show that the Noether stress-energy tensor for the matter sector, for Φ fields, is:

$$T^{\mu\nu}_{mat} = D^\mu\Phi^* D^\nu\Phi + D^\nu\Phi^* D^\mu\Phi - \eta^{\mu\nu}(D_\mu\Phi^* D^\mu\Phi - m^2\Phi^*\Phi)$$

d) 10 marks: Use the scalar fields equations of motion and the non-commutativity of covariant derivatives

$$[D_\mu, D_\nu]\Phi = ieF_{\mu\nu}\Phi, \qquad [D_\mu, D_\nu]\Phi^* = -ieF_{\mu\nu}\Phi^*$$

to show that

$$\partial_\mu T^{\mu\nu}_{mat} = F^{\nu\lambda}J_\lambda$$

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2. Supersymmetry: 25 marks

Consider the Lagrangian (χ is a left-handed, Weyl, fermion; ϕ is a complex scalar field; F is a complex "auxiliary" scalar field):

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi + i\chi^+ \bar{\sigma} \cdot \partial \chi + F^* F \tag{1}$$

a) 10 marks: Show that this Lagrangian is invariant under following infinitesimal transformations up to total derivative

$$\delta\phi = 0 \tag{2}$$

$$\delta\chi = \epsilon F \tag{3}$$

$$\delta F = -i\epsilon^+ \bar{\sigma} \cdot \partial \chi \tag{4}$$

Calculate only in the first order for ϵ, drop second order terms in ϵ during calculation. Here ϵ is anti-commuting variable (anti-commutes with itself and with other fermions, like χ). $\sigma^\mu = (1, \sigma^a)$; $\quad \bar{\sigma}^\mu = (1, -\sigma^a)$ and $\sigma \cdot \partial = \sigma^\mu \partial_\mu, \quad \bar{\sigma} \cdot \partial = \bar{\sigma}^\mu \partial_\mu$.

b) 15 marks: Show that this Lagrangian is also invariant up to total derivatives under more complicated infinitesimal transformation:

$$\delta\phi = -i\epsilon^T \sigma^2 \chi \tag{5}$$

$$\delta\chi = \epsilon F + \sigma \cdot \partial\phi \sigma^2 \epsilon^* \tag{6}$$

$$\delta F = -i\epsilon^+ \bar{\sigma} \cdot \partial \chi \tag{7}$$

Calculate only in the first order for ϵ, drop second order terms in ϵ during calculation.

Such symmetry is called supersymmetry.

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Note 1: use following identities about Weyl fermions and Pauli matrices:

$$\sigma^2\sigma^a\sigma^2 = -(\sigma^a)^*; \qquad a = 1, 2, 3$$

$$\sigma^2\sigma^\mu\sigma^2 = (\bar{\sigma}^\mu)^*$$

$$\chi^T\epsilon = \chi^a\epsilon^a = -\epsilon^a\chi^a = -\epsilon^T\chi$$

$$\chi^T\sigma^2\epsilon = \chi^a\sigma^2_{ab}\epsilon^b = -\chi^a\sigma^2_{ba}\epsilon^b = \epsilon^b\sigma^2_{ba}\chi^a = \epsilon^T\sigma^2\chi$$

Note 2: freely integrate by parts in $\int \mathcal{L}$ since question asks to show the invariance up to total derivatives.

Note 3: Assuming that it is invariant under transformations from question a) the question b) is reduced to checking the symmetry only under

$$\delta\phi = -i\epsilon^T\sigma^2\chi$$

$$\delta\chi = +\sigma\cdot\partial\phi\sigma^2\epsilon^*$$

$$\delta F = 0$$

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3. Lorentz transformation in Coulomb gauge

a) 10 Marks. Explain that under infinitesimal Lorentz transformation with parameter ϵ Maxwell field A_μ in Coulomb gauge $\partial_i A_i = 0$ transforms as:

$$U(\epsilon)A^\mu(x)U^{-1}(\epsilon) = A^\mu(x') - \epsilon^{\mu\nu}A_\nu(x') + \frac{\partial\Lambda(x',\epsilon)}{\partial x'_\mu}$$

for some Λ.

b) 15 Marks. Find Λ

4. Four-point functions in $\frac{\lambda}{4!}\phi^4$ theory: 25 marks

a) 10 marks: Draw all diagrams with four external legs in ϕ^4 theory up to one loop (there is one diagram in zero-loop and three diagrams in one-loop):

$$i\mathcal{M}(p_1p_2 \to p_3p_4) = -i\lambda + (-i\lambda)^2[iV(s) + iV(t) + iV(u)]$$

Write the expression for $V(s)$ as an integral over Feynman parameter x and loop four-momentum.

b) 15 marks: Use dimensional regularization to evaluate momentum integral (you don't have to integrate over Feynman parameter). $s = (p_1 + p_2)^2$.

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5. 2d QED: 25 Marks

a) 15 marks: Write down the expression for one loop photon two-point function in d-dimensional QED (Feynman rules are the same as in 4d, but all momentum integrals are d-dimensional). Introduce the Feynman parameter and simplify the expression.

b) 10 marks: Reduce it to the integral over Feynman parameter - evaluate the momentum integral after Euclidean rotation (integral over Feynman parameter is not necessary to compute) and express it through Gamma function. As opposed to 4d QED this one-loop diagram is convergent for d=2. Explain this from final expression.

6. Loop integrals: 25 marks

a) 5 marks: Show that for $m > 2$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{[l_E^2 + \Delta]^m} = \frac{1}{(4\pi)^2} \frac{1}{(m-1)(m-2)} \frac{1}{\Delta^{m-2}}$$

b) 5 marks Show that for $m > 3$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{l_E^2}{[l_E^2 + \Delta]^m} = \frac{2}{(4\pi)^2} \frac{1}{(m-1)(m-2)(m-3)} \frac{1}{\Delta^{m-3}}$$

c) 10 marks: Prove that for $n \geq 2$:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{[l_E^2 + \Delta]^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} (\frac{1}{\Delta})^{n - \frac{d}{2}}$$

d) 5 marks: Prove that for $n \geq 2$:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{l_E^2}{[l_E^2 + \Delta]^n} = \frac{d}{2(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n - \frac{d}{2} - 1)}{\Gamma(n)} (\frac{1}{\Delta})^{n - \frac{d}{2} - 1}$$

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